LAB – 2

1. DFS

```
Solution found!
Moves: 19
Step 0
(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

Step 1
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

Step 2
(1, 0, 3)
(4, 2, 6)
(7, 5, 8)

Step 3
(0, 1, 3)
(4, 2, 6)
(7, 5, 8)

Step 4
(4, 1, 3)
(0, 2, 6)
(7, 5, 8)

Step 5
(4, 1, 3)
(2, 0, 6)
(7, 5, 8)

Step 6
(4, 0, 3)
(2, 1, 6)
(7, 5, 8)
```

```
Step 7
(0, 4, 3)
(2, 1, 6)
(7, 5, 8)

Step 8
(2, 4, 3)
(0, 1, 6)
(7, 5, 8)

Step 9
(2, 4, 3)
(1, 0, 6)
(7, 5, 8)

Step 10
(2, 4, 3)
(1, 5, 6)
(7, 0, 8)

Step 11
(2, 4, 3)
(1, 5, 6)
(7, 8, 0)

Step 12
(2, 4, 3)
(1, 5, 0)
(7, 8, 6)

Step 13
(2, 4, 3)
(1, 0, 5)
(7, 8, 6)
```

```
Step 14
(2, 0, 3)
(1, 4, 5)
(7, 8, 6)

Step 15
(0, 2, 3)
(1, 4, 5)
(7, 8, 6)

Step 16
(1, 2, 3)
(0, 4, 5)
(7, 8, 6)

Step 17
(1, 2, 3)
(4, 0, 5)
(7, 8, 6)

Step 18
(1, 2, 3)
(4, 5, 0)
(7, 8, 6)

Step 19
(1, 2, 3)
(4, 5, 6)
(7, 8, 0)
```

2. IDS

```
Depth limit: 0
Depth limit: 1
Solution found!
Moves: 1
Step 0
(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

Step 1
(1, 2, 3)
(4, 5, 6)
(7, 8, 0)
```